

NeuronFabric 334K – Shakespeare FPGA Config

embed=88 heads=4 ff=264 layers=4 vocab=256

INPUT

token [0..255] → Embedding SRAM
vocab × embed = 256 × 88 [88 KB]

TRANSFORMER CORE (× 4 identical layers)

LayerNorm (88)
Attention: 4 heads × headDim=22
Wq Wk Wv: [88×22] Wo: [22×88] fan-in: 88
+ residual
LayerNorm (88)
FF: W1 [88 → 264] W2 [264 → 88] fan-in: 264
+ residual
Adam state per weight: m (FP32) + v (FP32)

× 4

OUTPUT

logits = layerOut · Embedding.T (weight-tied, no extra params)

Target: Xilinx ZCU102 BRAM = 4.0 MB | 334K BF16W = 3.34 MB ✓ (660 KB headroom)

SRAM Budget 334K params, ZCU102

